

MoMo payments through the Nightgals app: approved, declined, expired and failed

Viewers bought 1,000 in credit with MoMo through the Nightgals API, once for each of MTN's sandbox test numbers. The app sent MTN a callback address, MTN reported each result to it, and the app confirmed the status with MTN before completing or failing the purchase. Below is what the app logged, what MTN answered, the callbacks MTN sent, and what was stored. Keys and tokens are masked by the app itself.

Scenarios tested 6 payments	Approved 2 · credit added	Failed 4 · no credit added
Approved payment ● COMPLETED	MTN transaction ID 281824661	Pay → completed 3.3 s, by callback

APPROVED PAYMENT · 237689686224

What happened, in order

Times are UTC. The offset below each time counts from the buyer pressing "pay".

■ Buyer's app → Nightgals API
■ Nightgals → MTN
■ MTN → Nightgals
■ Nightgals updates the purchase

08:47:35.768
+0.0 s

BUYER → NIGHTGALS Buyer pays for 1,000 credit with MoMo

POST /api/v1/billing/credit/top-up. The app creates a PENDING purchase and uses its ID for everything that follows.

```
{
  "amountMinor": 1000,
  "method": "MOMO",
  "payerMsisdn": "237689686224"
}
```

08:47:35.924
+0.2 s

NIGHTGALS → MTN Gets an access token

MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/token/ headers=[Authorization:"Basic ****", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>

08:47:36.873
+1.1 s

MTN → NIGHTGALS Token issued, valid 1 hour 948 ms

MoMo ← 200 /collection/token/ 948ms body={
 "access_token": "****",
 "token_type": "access_token",
 "expires_in": 3600
}

08:47:36.914
+1.1 s

NIGHTGALS → MTN Sends the collection request with a callback address

X-Callback-Url: <https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo> tells MTN where to report the result. X-Reference-Id and externalId are both the purchase ID, so a retried checkout can't send a second prompt.

```
MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"221"] body={"payer":{"partyId":"237689686224","partyIdType":"MSISDN"},"externalId":"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f","amount":"1000","currency":"EUR","payeeNote":"Nightgals credit topup","payerMessage":"Nightgals credit topup"}
```

08:47:37.263
+1.5 s

MTN → NIGHTGALS Accepted: prompt sent to the handset 349 ms

202 means accepted, not paid. There is no body.

```
MoMo ← 202 /collection/v1_0/requesttopay 349ms body=<empty>
```

08:47:37.358
+1.6 s

NIGHTGALS Purchase left PENDING, buyer told to check their phone ● PENDING 1.6 s

MoMo prompt sent for purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f to ****6224

```
{
  "purchase": {
    "id": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
    "type": "CREDIT_TOPUP",
    "amountMinor": 1000,
    "priceDisplay": "1000",
    "currency": "XAF",
    "status": "PENDING",
    "provider": "MOMO",
    "createdAt": "2026-09-14T08:47:35.877722450Z"
  },
  "action": "PROMPT_ON_PHONE",
  "instructions": "Check your phone and approve the Mobile Money request."
}
```

08:47:38
+2.2 s

MTN → NIGHTGALS MTN calls the app back • SUCCESSFUL

POST /api/v1/webhooks/momo arrived through the tunnel within 2 s of MTN accepting the request, and the app answered 200. ngrok records this time only to the second.

```
{
  "financialTransactionId": "281824661",
  "externalId": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "237689686224"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "SUCCESSFUL"
}
```

08:47:38.715
+3.0 s

NIGHTGALS → MTN App confirms the status with MTN

The callback is unauthenticated, so the app never completes a purchase from it directly. It asks MTN over an authenticated call first.

```
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
```

08:47:39.027
+3.3 s

MTN → NIGHTGALS MTN confirms: approved • SUCCESSFUL 310 ms

```
{
  "financialTransactionId": "281824661",
  "externalId": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "237689686224"
  },
  "payerMessage": "Nightgals credit topup",
  "payeeNote": "Nightgals credit topup",
  "status": "SUCCESSFUL"
}
```

08:47:39.074
+3.3 s

NIGHTGALS Purchase completed and credit added • COMPLETED

Completed with MTN's transaction ID as the payment reference, then 1,000 added to the viewer's balance, 3.3 s after the buyer paid.

```
XAF 1,000 credited to a31049fe-cc72-496a-9283-c5a9136ef529 from top-up e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f
Purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled (CREDIT_TOPUP)
MoMo purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled by callback
```

08:47:52.401 ■
+16.6 s

BUYER → NIGHTGALS Buyer's app sees COMPLETED; balance is 1,000

● COMPLETED

The buyer's app polls every 15 s, so it saw the change on its next check.

```
{
  "purchase": {
    "id": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
    "type": "CREDIT_TOPUP",
    "amountMinor": 1000,
    "priceDisplay": "1000",
    "currency": "XAF",
    "status": "COMPLETED",
    "provider": "MOMO",
    "completedAt": "2026-09-14T08:47:39.034139Z",
    "createdAt": "2026-09-14T08:47:35.877722Z"
  },
  "credit": {
    "balanceMinor": 1000,
    "currency": "XAF"
  }
}
```

The callback MTN sent, as received

Captured by ngrok before it reached the app. It carries no signature or credential, which is why the app asks MTN for the status before completing anything.

```
POST /api/v1/webhooks/momo → HTTP 200
Content-Type: application/json; charset=utf-8
Content-Length: 240
User-Agent: LWAC Http Client 1.0
Host: unneighborly-keturah-noncontingent.ngrok-free.dev:443
X-Forwarded-For: 212.88.125.17
X-Forwarded-Proto: https
```

```
{
  "financialTransactionId": "281824661",
  "externalId": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "237689686224"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "SUCCESSFUL"
}
```

EVERY OUTCOME • MTN SANDBOX TEST NUMBERS

When the payment isn't approved straight away

The same checkout, run once for each of MTN's other test numbers. Each number makes MTN answer a different way. "Seen by status check" lists what MTN said when the app asked; "Callback" is what MTN reported on its own.

Payer number	Simulates	Seen by status check	Callback from MTN	Final purchase status	Pay → final	Buyer's credit after
46733123451	Payer declines	not asked; the callback came first	● FAILED APPROVAL_REJECTED	● FAILED APPROVAL_REJECTED	4.3 s · by callback	0 XAF
46733123452	Prompt expires	not asked; the callback came first	● FAILED EXPIRED	● FAILED EXPIRED	4.3 s · by callback	0 XAF
46733123450	MTN-side error	not asked; the callback came first	● FAILED INTERNAL_PROCESSING_ERROR	● FAILED INTERNAL_PROCESSING_ERROR	1.5 s · by callback	0 XAF
46733123453	No answer for a few minutes, then expires	● PENDING ● PENDING	● FAILED EXPIRED	● FAILED EXPIRED	3 min 17 s · by callback	0 XAF
46733123454	Approved after about a minute	● CREATED	● SUCCESSFUL	● COMPLETED	1 min 3 s · by callback	1000 XAF

46733123451 · Payer declines ● FAILED APPROVAL_REJECTED

MTN called back with FAILED (APPROVAL_REJECTED). The app asked MTN to confirm, then marked the purchase FAILED 4.3 s after the buyer paid. No credit was added.

```
MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"e0d594e8-8b8e-41c0-86ab-76db7bfc52ca", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"220"] body={
  "amount": "1000",
  "currency": "EUR",
  "payeeNote": "Nightgals credit topup",
  "payerMessage": "Nightgals credit topup",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "46733123451",
    "externalId": "e0d594e8-8b8e-41c0-86ab-76db7bfc52ca"
  }
}
MoMo prompt sent for purchase e0d594e8-8b8e-41c0-86ab-76db7bfc52ca to ****3451
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/e0d594e8-8b8e-41c0-86ab-76db7bfc52ca headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/e0d594e8-8b8e-41c0-86ab-76db7bfc52ca 336ms body={
  "externalId": "e0d594e8-8b8e-41c0-86ab-76db7bfc52ca",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "46733123451",
    "payerMessage": "Nightgals credit topup",
    "payeeNote": "Nightgals credit topup",
    "status": "FAILED",
    "reason": "APPROVAL_REJECTED"
  }
}
MoMo purchase e0d594e8-8b8e-41c0-86ab-76db7bfc52ca failed by callback: APPROVAL_REJECTED

{
  "externalId": "e0d594e8-8b8e-41c0-86ab-76db7bfc52ca",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "46733123451"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "FAILED",
  "reason": "APPROVAL_REJECTED"
}
```

46733123452 · Prompt expires ● **FAILED** EXPIRED

MTN called back with FAILED (EXPIRED). The app asked MTN to confirm, then marked the purchase FAILED 4.3 s after the buyer paid. No credit was added.

```
MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"d1b72e22-3beb-460f-ae9-7cde61939dfc", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"220"] body={
  "amount": "1000", "currency": "EUR", "payeeNote": "Nightgals credit topup", "payerMessage": "Nightgals credit topup", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123452", "externalId": "d1b72e22-3beb-460f-ae9-7cde61939dfc"
  }
}
MoMo prompt sent for purchase d1b72e22-3beb-460f-ae9-7cde61939dfc to ****3452
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/d1b72e22-3beb-460f-ae9-7cde61939dfc headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/d1b72e22-3beb-460f-ae9-7cde61939dfc 316ms body={
  "externalId": "d1b72e22-3beb-460f-ae9-7cde61939dfc", "amount": "1000", "currency": "EUR", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123452", "payerMessage": "Nightgals credit topup", "payeeNote": "Nightgals credit topup", "status": "FAILED", "reason": "EXPIRED"
  }
}
MoMo purchase d1b72e22-3beb-460f-ae9-7cde61939dfc failed by callback: EXPIRED
```

```
{
  "externalId": "d1b72e22-3beb-460f-ae9-7cde61939dfc",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "46733123452"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "FAILED",
  "reason": "EXPIRED"
}
```

46733123450 · MTN-side error ● **FAILED** INTERNAL_PROCESSING_ERROR

MTN called back with FAILED (INTERNAL_PROCESSING_ERROR). The app asked MTN to confirm, then marked the purchase FAILED 1.5 s after the buyer paid. No credit was added.

```
MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"220"] body={
  "amount": "1000", "currency": "EUR", "payeeNote": "Nightgals credit topup", "payerMessage": "Nightgals credit topup", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123450", "externalId": "b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376"
  }
}
MoMo prompt sent for purchase b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376 to ****3450
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376 headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376 311ms body={
  "externalId": "b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376", "amount": "1000", "currency": "EUR", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123450", "payerMessage": "Nightgals credit topup", "payeeNote": "Nightgals credit topup", "status": "FAILED", "reason": "INTERNAL_PROCESSING_ERROR"
  }
}
MoMo purchase b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376 failed by callback: INTERNAL_PROCESSING_ERROR
```

```
{
  "externalId": "b5abfa8b-8ae7-40b3-a349-3e8fbb3e4376",
  "amount": "1000",
  "currency": "EUR",
```

```

"payer": {
  "partyIdType": "MSISDN",
  "partyId": "46733123450"
},
"payeeNote": "Nightgals credit topup",
"status": "FAILED",
"reason": "INTERNAL_PROCESSING_ERROR"
}

```

46733123453 · No answer for a few minutes, then expires ● FAILED EXPIRED

The status checks saw PENDING, then PENDING and left the purchase PENDING. MTN called back with FAILED (EXPIRED). The app asked MTN to confirm, then marked the purchase FAILED 3 min 17 s after the buyer paid. No credit was added.

```

MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"74903e30-ad4f-4263-be1d-554b095258a4", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"220"] body=
{"amount":"1000","currency":"EUR","payeeNote":"Nightgals credit topup","payerMessage":"Nightgals credit topup","payer":{"partyIdType":"MSISDN","partyId":"46733123453"},"externalId":"74903e30-ad4f-4263-be1d-554b095258a4"}
MoMo prompt sent for purchase 74903e30-ad4f-4263-be1d-554b095258a4 to ****3453
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/74903e30-ad4f-4263-be1d-554b095258a4 headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/74903e30-ad4f-4263-be1d-554b095258a4 728ms body=
{"externalId":"74903e30-ad4f-4263-be1d-554b095258a4","amount":"1000","currency":"EUR","payer":{"partyIdType":"MSISDN","partyId":"46733123453"},"payerMessage":"Nightgals credit topup","payeeNote":"Nightgals credit topup","status":"PENDING"}
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/74903e30-ad4f-4263-be1d-554b095258a4 headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/74903e30-ad4f-4263-be1d-554b095258a4 1133ms body=
{"externalId":"74903e30-ad4f-4263-be1d-554b095258a4","amount":"1000","currency":"EUR","payer":{"partyIdType":"MSISDN","partyId":"46733123453"},"payerMessage":"Nightgals credit topup","payeeNote":"Nightgals credit topup","status":"PENDING"}
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/74903e30-ad4f-4263-be1d-554b095258a4 headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/74903e30-ad4f-4263-be1d-554b095258a4 314ms body=
{"externalId":"74903e30-ad4f-4263-be1d-554b095258a4","amount":"1000","currency":"EUR","payer":{"partyIdType":"MSISDN","partyId":"46733123453"},"payerMessage":"Nightgals credit topup","payeeNote":"Nightgals credit topup","status":"FAILED","reason":"EXPIRED"}
MoMo purchase 74903e30-ad4f-4263-be1d-554b095258a4 failed by callback: EXPIRED

```

```

{
  "externalId": "74903e30-ad4f-4263-be1d-554b095258a4",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "46733123453"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "FAILED",
  "reason": "EXPIRED"
}

```

46733123454 · Approved after about a minute ● COMPLETED

The status check saw CREATED and left the purchase PENDING. MTN called back with SUCCESSFUL. The app confirmed with MTN, completed the purchase and added 1,000 credit 1 min 3 s after the buyer paid.

```
MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"d5a53dee-cb80-483b-b84e-537a5e4ca426", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"220"] body={
  "amount": "1000", "currency": "EUR", "payeeNote": "Nightgals credit topup", "payerMessage": "Nightgals credit topup", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123454", "externalId": "d5a53dee-cb80-483b-b84e-537a5e4ca426"
  }
}
MoMo prompt sent for purchase d5a53dee-cb80-483b-b84e-537a5e4ca426 to ****3454
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/d5a53dee-cb80-483b-b84e-537a5e4ca426 headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/d5a53dee-cb80-483b-b84e-537a5e4ca426 309ms body={
  "externalId": "d5a53dee-cb80-483b-b84e-537a5e4ca426", "amount": "1000", "currency": "EUR", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123454", "payerMessage": "Nightgals credit topup", "payeeNote": "Nightgals credit topup", "status": "CREATED"
  }
}
MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/d5a53dee-cb80-483b-b84e-537a5e4ca426 headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
MoMo ← 200 /collection/v1_0/requesttopay/d5a53dee-cb80-483b-b84e-537a5e4ca426 349ms body={
  "financialTransactionId": "1523335279", "externalId": "d5a53dee-cb80-483b-b84e-537a5e4ca426", "amount": "1000", "currency": "EUR", "payer": {
    "partyIdType": "MSISDN", "partyId": "46733123454", "payerMessage": "Nightgals credit topup", "payeeNote": "Nightgals credit topup", "status": "SUCCESSFUL"
  }
}
XAF 1,000 credited to 4117043f-5e7e-4e97-873a-0a58e2dd9c09 from top-up d5a53dee-cb80-483b-b84e-537a5e4ca426
Purchase d5a53dee-cb80-483b-b84e-537a5e4ca426 settled (CREDIT_TOPUP)
MoMo purchase d5a53dee-cb80-483b-b84e-537a5e4ca426 settled by callback
```

```
{
  "financialTransactionId": "1523335279",
  "externalId": "d5a53dee-cb80-483b-b84e-537a5e4ca426",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "46733123454"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "SUCCESSFUL"
}
```

RESILIENCE

Calls to MTN can't hang

Every call to MTN now has a time limit, for connecting and again for the answer. It's set by MOMO_TIMEOUT and defaults to 10 seconds.

What happens when MTN doesn't answer in time

At checkout: the buyer gets "Mobile Money is not responding. Try again in a moment." A retry reuses the same purchase ID as the reference, so it can't send a second prompt.

During a status check: that purchase is treated as "no answer yet" and asked about again on the next check. The other pending purchases in the same check are unaffected.

When a callback arrives: the app still answers MTN with 200, and the purchase is picked up by the next status check.

Automated test: an MTN that accepts the connection and never answers

Run against a local server that holds the connection open without replying, with the time limit set to 1 second.

Test	Expected	Result
Status check against a silent MTN	Returns "no answer" within 5 s instead of hanging	● PASSED
Payment request against a silent MTN	Reports MTN unavailable within 5 s instead of hanging	● PASSED

```
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.567 s -- in
com.nightgals.MomoClientTimeoutTest
WARN  MomoClient -- MoMo status fcd23965-... unavailable: I/O error on POST request for
"http://127.0.0.1:41749/collection/token/": Request cancelled
ERROR MomoClient -- MoMo requesttopay b21f7641-... failed: I/O error on POST request for
"http://127.0.0.1:45393/collection/token/": Request cancelled
```

What we found

The callback flow works end to end

MTN posted the result to /api/v1/webhooks/momo within 2 s of accepting the request. The app confirmed the status with MTN, completed the purchase with MTN's transaction ID as the payment reference, and added 1,000 to the viewer's balance, 3.3 s after the buyer paid. One purchase completed, one credit entry.

Declined, expired and failed payments never give access

For every failure, MTN called back within seconds of its final answer. The app confirmed the status with MTN, marked the purchase FAILED with MTN's reason (APPROVAL_REJECTED, EXPIRED or INTERNAL_PROCESSING_ERROR), and added no credit. A payment MTN hadn't decided yet was left PENDING at each status check until MTN gave its final answer.

The app records XAF, while sandbox is charged EUR

The purchase is stored as 1,000 XAF, but the request to MTN says "currency": "EUR" because MOMO_CURRENCY=EUR. That's expected: sandbox only accepts EUR. The currency and target environment must switch to XAF and the live values at the same time.

The app's log

Every MoMo and billing line the running app wrote during the approved payment, exactly as logged; each failure's lines are in its own entry above (timestamps are the machine's local time, UTC+3). The colored edge marks requests to MTN, MTN's answers and purchase updates.

10 lines · nightgals-app4.log, filtered to MoMo and billing

```
11:47:35.924 DEBUG [omcat-handler-2] MomoClient MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/token/ headers=[Authorization:"Basic ****", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
11:47:36.873 DEBUG [omcat-handler-2] MomoClient MoMo ← 200 /collection/token/ 948ms body={
  "access_token":"****",
  "token_type":"access_token",
  "expires_in":3600}
11:47:36.914 DEBUG [omcat-handler-2] MomoClient MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"221"] body={
  "payer":{
    "partyId":"237689686224",
    "partyIdType":"MSISDN"},
    "externalId":"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
    "amount":"1000",
    "currency":"EUR",
    "payeeNote":"Nightgals credit topup",
    "payerMessage":"Nightgals credit topup"}
11:47:37.263 DEBUG [omcat-handler-2] MomoClient MoMo ← 202 /collection/v1_0/requesttopay 349ms body=<empty>
11:47:37.264 INFO [omcat-handler-2] MomoPaymentProvider MoMo prompt sent for purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f to ****6224
11:47:38.715 DEBUG [omcat-handler-4] MomoClient MoMo → GET https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
11:47:39.027 DEBUG [omcat-handler-4] MomoClient MoMo ← 200 /collection/v1_0/requesttopay/e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f 310ms body={
  "financialTransactionId":"281824661",
  "externalId":"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount":"1000",
  "currency":"EUR",
  "payer":{
    "partyIdType":"MSISDN",
    "partyId":"237689686224"},
    "payerMessage":"Nightgals credit topup",
    "payeeNote":"Nightgals credit topup",
    "status":"SUCCESSFUL"}
11:47:39.073 INFO [omcat-handler-4] CreditService XAF 1,000 credited to a31049fe-cc72-496a-9283-c5a9136ef529 from top-up e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f
11:47:39.074 INFO [omcat-handler-4] BillingService Purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled (CREDIT_TOPUP)
11:47:39.088 INFO [omcat-handler-4] MomoCallbackController MoMo purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled by callback
```

The buyer's API calls for the approved payment

What a mobile client would send. The purchase was polled every 15 seconds until it changed.

UTC	Call	HTTP	Took	Purchase status
08:47:30.112	POST /api/v1/auth/register	201	5.6 s	
08:47:35.744	GET /api/v1/billing/payment-methods	200	24 ms	
08:47:35.768	POST /api/v1/billing/credit/top-up	200	1.6 s	● PENDING

UTC	Call	HTTP	Took	Purchase status
08:47:37.358	GET /api/v1/billing/purchases/{purchaseId}	200	42 ms	● PENDING
08:47:52.401	GET /api/v1/billing/purchases/{purchaseId}	200	15 ms	● COMPLETED
08:47:52.416	GET /api/v1/billing/credit	200	21 ms	

Stored afterwards

Read from the local Postgres after the run.

id	e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f
type	CREDIT_TOPUP
status	COMPLETED
provider	MOMO
amount_minor	1000
currency	XAF
payer_msisdn	237689686224
provider_reference	281824661
created_at	2026-09-14 08:47:35.877 UTC
completed_at	2026-09-14 08:47:39.034 UTC
credit entries	1 · TOPUP · 1000 XAF · 08:47:39.054 UTC

How this was run

- An ngrok tunnel gave the local app a public HTTPS address, and MOMO_CALLBACK_URL was set to its /api/v1/webhooks/momo address.
 - A sandbox API user registered to the tunnel’s host, since MTN only calls back to the registered host.
 - Buyers: a new viewer account (momo-test-...@example.com) for each payment. Payers: MTN’s sandbox test numbers, each of which makes MTN answer a fixed way.
 - For the failure runs, an unanswered payment was set to be given up after 3 minutes instead of 1 hour, so a stuck payment would show up during the test. MTN gave a final answer for every payment before that limit, so it wasn’t reached.

Run on 2026-09-14 against sandbox.momodeveloper.mtn.com from a local Nightgals build. The buyer accounts were created for these tests only. Keys, tokens and the account password never appear here: the app masks MoMo credentials in its own log.